tf.bitwise.left_shift Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/bitwise/left_shift

> *Elementwise computes the bitwise left-shift of x and y.*

Function Signatures

```
tf.bitwise.left_shift( x: Annotated[Any, TV_LeftShift_T], y: Annotated[Any, TV_LeftShift_T], name=None ) -> Annotated[Any, TV_LeftShift_T]
```

Code Examples

```
tf.bitwise.left_shift(
x: Annotated[Any, TV_LeftShift_T],
y: Annotated[Any, TV_LeftShift_T],
name=None
) -> Annotated[Any, TV_LeftShift_T]
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x: Annotated[Any, TV_LeftShift_T],
y: Annotated[Any, TV_LeftShift_T],
name=None
) -> Annotated[Any, TV_LeftShift_T]
```

```
import tensorflow as tf
from tensorflow.python.ops import bitwise_ops
import numpy as np
dtype_list = [tf.int8, tf.int16, tf.int32, tf.int64]
for dtype in dtype_list:
lhs = tf.constant([-1, -5, -3, -14], dtype=dtype)
rhs = tf.constant([5, 0, 7, 11], dtype=dtype)
left_shift_result = bitwise_ops.left_shift(lhs, rhs)
print(left_shift_result)
# This will print:
# tf.Tensor([ -32   -5 -128    0], shape=(4,), dtype=int8)
# tf.Tensor([   -32     -5   -384 -28672], shape=(4,), dtype=int16)
# tf.Tensor([   -32     -5   -384 -28672], shape=(4,), dtype=int32)
# tf.Tensor([   -32     -5   -384 -28672], shape=(4,), dtype=int64)
lhs = np.array([-2, 64, 101, 32], dtype=np.int8)
rhs = np.array([-1, -5, -3, -14], dtype=np.int8)
bitwise_ops.left_shift(lhs, rhs)
#
```


```
import tensorflow as tf
from tensorflow.python.ops import bitwise_ops
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dtype_list = [tf.int8, tf.int16, tf.int32, tf.int64]
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# tf.Tensor([   -32     -5   -384 -28672], shape=(4,), dtype=int32)
# tf.Tensor([   -32     -5   -384 -28672], shape=(4,), dtype=int64)
lhs = np.array([-2, 64, 101, 32], dtype=np.int8)
rhs = np.array([-1, -5, -3, -14], dtype=np.int8)
bitwise_ops.left_shift(lhs, rhs)
#
```

Documentation

Elementwise computes the bitwise left-shift of x and y.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.bitwise.left_shift

If y is negative, or greater than or equal to the width of x in bits the result is implementation defined.

Example:

Returns